

Problem

Obtain the exponential Fourier series for $f(t) = t$, $-1 < t < 1$, with $f(t + 2n) = f(t)$ for all integer values of n .

Step-by-step solution

Step 1 of 4

The function is,

$$f(t) = t \quad \text{for } -1 < t < 1$$

The time period is,

$$T = 2$$

Therefore,

$$\begin{aligned}\omega_0 &= \frac{2\pi}{T} \\ &= \frac{2\pi}{2} \\ &= \pi\end{aligned}$$

Step 2 of 4

Calculate the coefficient of exponential Fourier series.

$$C_n = \frac{1}{T} \int_0^T f(t) e^{-jn\omega_0 t} dt$$

$$C_n = \frac{1}{2} \int_{-1}^1 t e^{-jn\pi t} dt$$

Using integration by parts,

$$u = t$$

$$du = dt$$

And,

$$dv = e^{-jn\pi t}$$

Apply integration on both sides.

$$\int dv = \int e^{-jn\pi t}$$

$$v = -\frac{1}{jn\pi} e^{-jn\pi t}$$

Step 3 of 4

Therefore,

$$\begin{aligned}
 C_n &= -\frac{t}{2jn\pi} e^{-jn\pi t} \Big|_{-1}^1 + \frac{1}{2jn\pi} \int_{-1}^1 e^{-jn\pi t} dt \\
 &= -\frac{t}{2jn\pi} e^{-jn\pi t} \Big|_{-1}^1 + \frac{1}{2jn\pi} \left[\frac{e^{-jn\pi t}}{-jn\pi} \right]_{-1}^1 \\
 &= -\left(\frac{1}{2jn\pi} e^{-jn\pi} - \frac{-1}{2jn\pi} e^{jn\pi} \right) + \frac{1}{2n^2\pi^2} [e^{-jn\pi} - e^{jn\pi}] \\
 &= -\frac{1}{2jn\pi} (e^{-jn\pi} + e^{jn\pi}) + \frac{1}{2n^2\pi^2} [e^{-jn\pi} - e^{jn\pi}] \\
 &= \frac{j}{2n\pi} 2 \cos(n\pi) + \frac{2j}{2n^2\pi^2} \sin(n\pi) \\
 &= \frac{j(-1)^n}{n\pi} + \frac{2j}{2n^2\pi^2} (0) \\
 &= \frac{j(-1)^n}{n\pi}
 \end{aligned}$$

Step 4 of 4

The exponential Fourier series of $f(t)$ is,

$$\begin{aligned}
 f(t) &= \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t} \\
 &= \sum_{n=-\infty}^{\infty} \frac{j(-1)^n}{n\pi} e^{jn\pi t} \\
 &= \sum_{n=-\infty}^{\infty} (-1)^n \frac{j}{n\pi} e^{jn\pi t}
 \end{aligned}$$

Therefore, the exponential Fourier series, $f(t)$ is $\boxed{\sum_{n=-\infty}^{\infty} (-1)^n \frac{j}{n\pi} e^{jn\pi t}}$.